

Numerical Linear Algebra - Tutorial

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Exercise 1

Let

$$u = \begin{pmatrix} -1 \\ 2 \end{pmatrix}, \quad v = \begin{pmatrix} 4 \\ 6 \end{pmatrix}, \quad w = \begin{pmatrix} 3 \\ -1 \\ -5 \end{pmatrix}, \quad x = \begin{pmatrix} 6 \\ -2 \\ 3 \end{pmatrix}$$

Compute:

- (i) $\frac{1}{w \cdot w} w$
- (ii) $\left(\frac{x \cdot w}{x \cdot x} \right) x$
- (iii) $\left(\frac{u \cdot v}{v \cdot v} \right) v$
- (iv) $\|w\|$
- (v) $\|x\|$

Exercise 2

Find the distance between

$$u = \begin{pmatrix} 0 \\ -5 \\ 2 \end{pmatrix}, \quad z = \begin{pmatrix} -4 \\ -1 \\ 8 \end{pmatrix}.$$

Exercise 3

State whether the following statements are **True or False**. Justify your answer.

- (a) $v \cdot v = \|v\|^2$
- (b) For any scalar c , $u \cdot (cv) = c(u \cdot v)$
- (c) If the distance from u to v equals that from u to $-v$, then u and v are orthogonal.
- (d) For a square matrix A , vectors in $\text{Col}(A)$ are orthogonal to vectors in $\text{Null}(A)$.
- (e) If vectors $v_1, \dots, v_p$ span a subspace W and if x is orthogonal to each v_j , then $x \in W^\perp$.

Exercise 4

Let $v = \begin{pmatrix} a \\ b \end{pmatrix}$. Describe the set $H \subset \mathbb{R}^2$ of vectors $\begin{pmatrix} x \\ y \end{pmatrix}$ that are orthogonal to v .

Exercise 5

Show that $\{v_1, v_2, v_3\}$ is an orthogonal basis of $\mathbb{R}^3$, where

$$v_1 = \left(\frac{3}{\sqrt{11}}, \frac{1}{\sqrt{11}}, \frac{1}{\sqrt{11}} \right), \quad v_2 = \left(-\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}} \right), \quad v_3 = \left(-\frac{1}{\sqrt{66}}, -\frac{4}{\sqrt{66}}, \frac{7}{\sqrt{66}} \right)$$

Exercise 6

Prove that an $m \times n$ matrix U has orthogonal columns if and only if $U^T U = I$.

Exercise 7

Let $U \in \mathbb{R}^{m \times n}$ be a matrix with orthogonal columns. Show that for all $x, y \in \mathbb{R}^n$:

- (a) $\|Ux\| = \|x\|$
- (b) $(Ux) \cdot (Uy) = x \cdot y$
- (c) $(Ux) \cdot (Uy) = 0 \Leftrightarrow x \cdot y = 0$

Exercise 8

Let

$$u_1 = \begin{pmatrix} -7 \\ 1 \\ 4 \end{pmatrix}, \quad u_2 = \begin{pmatrix} -1 \\ 1 \\ -2 \end{pmatrix}, \quad y = \begin{pmatrix} -9 \\ 1 \\ 6 \end{pmatrix}, \quad w = \text{span}(u_1, u_2).$$

Find $\text{proj}_w(y)$.

Exercise 9

Let $x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$. Compute $x^T A x$ for:

- (a) $A = \begin{pmatrix} 4 & 0 \\ 0 & 3 \end{pmatrix}$
- (b) $A = \begin{pmatrix} 3 & -2 \\ -2 & 7 \end{pmatrix}$

Exercise 10

Let

$$Q(x) = 5x_1^2 + 3x_2^2 + 2x_3^2 - x_1x_2 + 8x_2x_3.$$

Write this quadratic form as $x^T A x$, with $x \in \mathbb{R}^3$.

Answer: $Q(x) = x^T A x = \begin{pmatrix} x_1 & x_2 & x_3 \end{pmatrix} \begin{pmatrix} 5 & -\frac{1}{2} & 0 \\ -\frac{1}{2} & 3 & 4 \\ 0 & 4 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$

Exercise 11 — Orthogonal Projection

Let $u = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ and $v = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$.

- Compute the orthogonal projection of v onto the line spanned by u (1D case).
- Compute the projection matrix P that projects any vector in $\mathbb{R}^2$ onto the line $\text{Span}(u)$, and verify that $Pv = \text{proj}_u(v)$.

Let now $u_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$, $u_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$, and $v = \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix}$.

- Compute the orthogonal projection of v onto the plane spanned by u_1 and u_2 (2D case).

Exercise 12 — Gram–Schmidt Orthogonalization

Let the vectors in $\mathbb{R}^3$ be:

$$v_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \quad v_3 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

Apply the Gram–Schmidt process to construct an orthonormal basis $\{u_1, u_2, u_3\}$ from $\{v_1, v_2, v_3\}$. Show all steps clearly:

- Start by computing $u_1 = \frac{v_1}{\|v_1\|}$
- Then subtract projections from v_2 and normalize to get u_2
- Finally, subtract the projections of v_3 onto both u_1 and u_2 to find u_3

Exercise 13 — Symmetric Matrix Decomposition

Let $A = \begin{pmatrix} 4 & 2 & 2 \\ 2 & 5 & 1 \\ 2 & 1 & 3 \end{pmatrix}$ be a symmetric matrix.

- Show that A is symmetric.
- Compute the eigenvalues and eigenvectors of A .
- Use your results to express A in the form $A = Q\Lambda Q^T$, where Q is an orthogonal matrix and Λ is diagonal (spectral decomposition).

Exercise 14 — Singular Value Decomposition (SVD)

- Let A be the 2×2 matrix:

$$A = \begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix}$$

Compute the singular value decomposition of A : that is, find orthogonal matrices U , V , and diagonal matrix Σ such that $A = U\Sigma V^T$.

- (b) Repeat the same task for the rectangular matrix:

$$B = \begin{pmatrix} 1 & 0 \\ 0 & 2 \\ 0 & 0 \end{pmatrix}$$

Find the SVD of B .

Exercise 15 — Low-Rank Approximation

Let

$$A = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

be a diagonal matrix of rank 3.

- Use the singular value decomposition of A to write a rank-1 approximation of A , denoted A_1 , by keeping only the largest singular value.
- Compute the approximation error in Frobenius norm: $\|A - A_1\|_F$.
- Explain how this approximation could be useful in data compression or dimensionality reduction.

Exercise 16 — Rayleigh Quotient and Symmetric Matrices

Let A be a real symmetric matrix and $\mathbf{x} \in \mathbb{R}^n \setminus \{0\}$. The Rayleigh quotient is defined as

$$R(\mathbf{x}) = \frac{\mathbf{x}^T A \mathbf{x}}{\mathbf{x}^T \mathbf{x}}.$$

- Show that if A is symmetric and $\mathbf{x}$ is an eigenvector of A , then $R(\mathbf{x})$ equals the corresponding eigenvalue.
- Let

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Compute the Rayleigh quotient $R(\mathbf{x})$. Compare the result with the eigenvalues of A .

- Explain why the Rayleigh quotient is useful for estimating the dominant eigenvalue of a symmetric matrix and how it relates to the power method.